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AI-Powered Fake News Detection Using Text Classification

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Abstract

The rapid growth of digital media platforms has significantly increased the accessibility of online news. Although this has improved the speed of information dissemination, it has also contributed to the widespread circulation of fake news, misleading information, and deliberately manipulated content. Fake news has become a serious global concern because it influences public opinion, political decisions, financial markets, and public health. Manual verification of news articles is time-consuming and impractical considering the enormous volume of information generated every day.

This project presents an Artificial Intelligence based fake news detection system using Natural Language Processing (NLP) and supervised machine learning techniques. The proposed system performs automatic classification of news articles into two categories: Fake News and Real News. The complete pipeline includes dataset preprocessing, feature engineering using Bag-of-Words, TF-IDF, and Word2Vec representations, followed by classification using Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and Multi-Layer Perceptron (MLP).

Experimental evaluation demonstrates that the Random Forest classifier achieved the highest overall performance among all evaluated models, achieving an accuracy of 99.79%, while Logistic Regression and Neural Networks also produced highly competitive results. The obtained results demonstrate that classical machine learning algorithms remain highly effective for fake news detection when combined with appropriate text preprocessing and feature extraction techniques.

Fake News Detection, Natural Language Processing, Machine Learning, Text Classification, TF-IDF, Bag of Words, Word2Vec, Random Forest, Logistic Regression, Artificial Intelligence

1 Introduction

The exponential growth of internet connectivity and social media platforms has transformed the way people consume news and information. Millions of news articles, blog posts, and social media updates are published every day across various online platforms. Although this digital transformation has made information more accessible than ever before, it has also introduced significant challenges related to information authenticity and credibility.

Fake news refers to intentionally fabricated or misleading information presented in the form of legitimate news with the objective of deceiving readers. Such misinformation often spreads rapidly because online platforms enable instantaneous sharing among millions of users. Unlike traditional journalism, fake news articles are usually designed to attract attention through sensational headlines, emotionally charged language, or manipulated facts.

The impact of fake news extends across multiple domains including politics, healthcare, finance, education, and disaster management. During elections, misinformation can influence voting behaviour and public perception. During public health emergencies, inaccurate medical information may discourage vaccination or promote unsafe treatments. Financial misinformation can manipulate stock prices and affect investor confidence. Consequently, developing automated fake news detection systems has become an active area of research within Artificial Intelligence and Natural Language Processing.

Traditional manual verification methods involve professional fact-checkers who compare published claims against trusted sources. Although manual verification provides high reliability, it is neither scalable nor capable of processing the enormous quantity of online content generated every minute. Therefore, automated approaches based on machine learning have emerged as practical solutions for identifying fake news with minimal human intervention.

Machine learning algorithms can learn discriminative textual patterns from previously labelled news articles. After suitable preprocessing and feature extraction, these algorithms classify unseen articles into fake or genuine categories based on learned statistical relationships. Recent advances in Natural Language Processing have further improved classification performance by transforming raw textual data into meaningful numerical representations suitable for machine learning models.

This project develops a complete fake news detection pipeline using supervised machine learning techniques. The proposed methodology begins by collecting labelled news articles from a publicly available dataset. The textual data is cleaned through multiple preprocessing stages including lowercase conversion, punctuation removal, URL removal, stop-word elimination, and tokenization. Feature extraction is then performed using three different approaches: Bag-of-Words (BoW), Term Frequency–Inverse Document Frequency (TF-IDF), and Word2Vec embeddings. Finally, four supervised learning algorithms are trained and evaluated to determine the most effective classifier for fake news detection.

The project demonstrates that careful preprocessing combined with appropriate feature engineering significantly improves classification performance. Experimental comparisons also provide insight into the strengths and weaknesses of parametric and non-parametric machine learning algorithms for text classification tasks.

2 Problem Statement

The widespread availability of social media has dramatically increased the speed at which misinformation spreads across the internet. Unlike verified journalism, fake news is often intentionally designed to manipulate public opinion by presenting fabricated or misleading information as authentic news. The enormous quantity of online content generated daily makes manual verification impossible on a large scale.

Therefore, there exists a need for an automated system capable of accurately distinguishing between fake and genuine news articles using machine learning techniques. Such a system should provide high classification accuracy while remaining computationally efficient and scalable for real-world applications.

3 Objectives

The primary objectives of this project are:

1. Develop an end-to-end fake news detection pipeline using Natural Language Processing.
2. Perform comprehensive text preprocessing to improve data quality.
3. Generate multiple feature representations using Bag-of-Words, TF-IDF, and Word2Vec.
4. Train and compare multiple supervised machine learning models.
5. Evaluate each model using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
6. Identify the most suitable classifier for fake news detection based on experimental performance.

4 Literature Review

Automatic fake news detection has become one of the most rapidly growing research areas within Artificial Intelligence and Natural Language Processing. Researchers have proposed numerous statistical, linguistic, and machine learning approaches for identifying misleading information in textual documents.

Early fake news detection systems primarily relied on handcrafted linguistic features such as writing style, grammatical structure, lexical diversity, sentiment analysis, and readability indices. Although these manually engineered features provided moderate classification accuracy, their effectiveness depended heavily on domain expertise and extensive feature engineering.

The emergence of supervised machine learning significantly improved fake news detection performance. Traditional classifiers including Naive Bayes, Support Vector Machines (SVM), Decision Trees, Logistic Regression, and Random

Forests demonstrated strong performance when combined with Bag-of-Words or TF-IDF representations. These approaches transformed textual documents into numerical vectors, enabling efficient statistical learning from labelled datasets.

Recent developments in word embedding techniques further enhanced semantic representation of textual information. Word2Vec, introduced by Mikolov et al., generates dense vector representations that capture semantic similarity between words. Unlike sparse Bag-of-Words vectors, Word2Vec embeddings preserve contextual relationships, thereby improving the learning capability of many machine learning algorithms. With the rapid advancement of deep learning, transformer-based architectures such as BERT, RoBERTa, DistilBERT, and XLNet have significantly improved fake news detection accuracy by capturing contextual semantics more effectively than traditional feature extraction techniques. These models leverage self-attention mechanisms to understand long-range dependencies within textual data. However, transformer models require substantial computational resources, large labelled datasets, and significantly longer training times.

Despite these advances, classical machine learning techniques remain attractive for academic and industrial applications because of their simplicity, lower computational requirements, faster training, and easier interpretability. Numerous studies have demonstrated that traditional classifiers such as Logistic Regression and Random Forest, when combined with TF-IDF features, achieve competitive performance on benchmark fake news datasets.

The present work focuses on evaluating four supervised machine learning algorithms using three different textual representations. Instead of relying on computationally intensive transformer architectures, the proposed system demonstrates that properly engineered classical machine learning pipelines can still achieve excellent classification performance while remaining computationally efficient.

5 Dataset Description

The dataset used in this project is the publicly available **Fake and Real News Dataset** obtained from Kaggle. It is one of the most widely used benchmark datasets for fake news detection research due to its balanced distribution of fake and genuine news articles.

The dataset consists of two comma-separated value (CSV) files:

- Fake.csv
- True.csv

The Fake.csv file contains fabricated news articles collected from unreliable news sources, whereas the True.csv file contains authentic news articles collected from verified media organizations.

Each news article contains textual and metadata information including title, article body, subject category, and publication date.

During preprocessing, both datasets were merged into a single dataset and binary class labels were assigned:

- Fake News = 0
- Real News = 1

After merging, the dataset was randomly shuffled to eliminate ordering bias before training.

5.1 Dataset Features

The original dataset contains the following attributes.

Table 1: Dataset Attributes

Feature	Description
Title	News headline
Text	Complete news article
Subject	News category
Date	Publication date
Label	Binary class (0 or 1)

5.2 Dataset Statistics

Table 2: Dataset Summary

Property	Value
Dataset Source	Kaggle
Language	English
Classification Type	Binary
Total Samples	44,898
Training Samples (80%)	35,918
Testing Samples (20%)	8,980
Classes	Fake, Real

6 Methodology

The overall workflow adopted in this project consists of multiple sequential stages beginning with dataset acquisition and ending with performance evaluation.

The complete pipeline is illustrated below.

Dataset → Preprocessing → Feature Engineering → Model Training → Prediction → Evaluation

Each stage contributes towards improving the overall classification accuracy.

6.1 Data Preprocessing

Raw textual data contains several unwanted elements such as punctuation symbols, hyperlinks, HTML tags, numbers, and common stopwords that negatively affect machine learning algorithms.

Therefore, multiple preprocessing operations were performed before feature extraction.

The preprocessing pipeline consists of:

1. Dataset merging
2. Label assignment
3. Random shuffling
4. Lowercase conversion
5. URL removal
6. HTML tag removal
7. Number removal
8. Punctuation removal
9. Whitespace normalization
10. Tokenization
11. Stopword removal
12. Clean dataset generation

6.2 Text Cleaning

Every news article was first converted to lowercase to eliminate duplicate words caused by capitalization.

URLs were removed using regular expression matching since hyperlinks contribute little towards identifying fake news.

HTML tags were removed to eliminate formatting information.

Numbers and punctuation symbols were discarded because they generally carry limited semantic meaning for textual classification.

Finally, multiple consecutive whitespaces were replaced with a single space to produce clean text suitable for tokenization.

6.3 Tokenization and Stopword Removal

The cleaned sentences were tokenized using the Natural Language Toolkit (NLTK).

Each sentence was divided into individual words.

Frequently occurring English stopwords such as

the, is, are, a, an, of, to

were removed because these words appear frequently across nearly all documents while contributing minimal discriminative information.

Removing stopwords reduces vocabulary size and improves computational efficiency.

6.4 Feature Engineering

Machine learning algorithms require numerical input.

Therefore, textual news articles were transformed into numerical vectors using three different feature extraction techniques.

- Bag of Words
- TF-IDF
- Word2Vec

6.5 Bag of Words

Bag-of-Words (BoW) is one of the simplest feature extraction techniques used in Natural Language Processing.

A vocabulary containing the most frequent words is first generated.

Each news article is then represented by a sparse vector where every element indicates the frequency of a particular word.

Although BoW ignores word ordering, it provides an effective representation for many document classification tasks.

6.6 TF-IDF

Term Frequency–Inverse Document Frequency (TF-IDF) extends Bag-of-Words by assigning larger weights to informative words while reducing the influence of extremely common words.

The TF-IDF score is computed as

TF(t,d)=Number of occurrences of term t $\frac{}{\text{Total number of terms in document } d}$
(1)

The inverse document frequency is

IDF(t)=$\log\left(\frac{N}{DF(t)}\right)$
(2)

where

- N = Total number of documents
- $DF(t)$ = Number of documents containing term t

Finally,

TFIDF(t,d)=TF(t,d)$\times IDF(t)$
(3)

This representation generally produces better classification performance because it emphasizes discriminative words.

6.7 Word2Vec

Word2Vec is a neural embedding technique that converts words into dense continuous vectors.

Unlike Bag-of-Words and TF-IDF, Word2Vec captures semantic similarity between words.

Words appearing in similar contexts are assigned similar vector representations.

In this project, a Word2Vec model was trained using the cleaned news articles.

The embedding vector for each news article was computed by averaging the vectors of all words contained within the article.

This dense numerical representation was later used as input for the K-Nearest Neighbors classifier.

7 Machine Learning Models

After transforming the textual news articles into numerical feature vectors, four supervised machine learning algorithms were trained and evaluated. These algorithms were selected because they represent different learning paradigms and are widely used in text classification tasks. The selected models include Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and Multi-Layer Perceptron (MLP). Their performances were compared using multiple evaluation metrics to determine the most suitable classifier for fake news detection.

7.1 Logistic Regression

Logistic Regression is a supervised parametric classification algorithm that estimates the probability of an input belonging to a particular class. Although originally developed for binary classification, it has become one of the most effective baseline algorithms for Natural Language Processing tasks due to its simplicity, interpretability, and computational efficiency.

Unlike linear regression, Logistic Regression employs the sigmoid activation function to transform a linear combination of input features into probabilities ranging between 0 and 1.

The sigmoid function is expressed as

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{4}$$

where

$$z = w^T x + b$$

Here,

- x represents the feature vector.
- w denotes the learned weight vector.
- b represents the bias term.

The model predicts the probability that a news article belongs to the Real News class. A threshold of 0.5 is generally used to classify the article into one of the two classes.

Logistic Regression performs exceptionally well on sparse high-dimensional textual data such as TF-IDF vectors because the learned decision boundary efficiently separates fake and genuine news articles.

Advantages

- Fast training
- Computationally efficient
- Easy to interpret
- Excellent for high-dimensional sparse data

Limitations

- Assumes linear separability
- Limited capability for modeling highly complex relationships

7.2 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a supervised non-parametric learning algorithm that classifies a sample based on the labels of its nearest neighboring observations in the feature space.

Instead of learning explicit parameters during training, KNN stores all training samples and performs classification during prediction.

The Euclidean distance between two feature vectors is computed as

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^{n} (x_i - x_j)^2} \tag{5}$$

The predicted class corresponds to the majority class among the nearest neighbors.

In this project, Word2Vec embeddings were used with KNN because dense semantic vectors provide meaningful distance measurements between documents.

Advantages

- Simple implementation
- No explicit training phase
- Effective on small datasets

Limitations

- Slow prediction
- High memory consumption
- Sensitive to irrelevant features

7.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to improve classification accuracy and reduce overfitting.

Each tree is trained using a random subset of the training data and a random subset of available features. Final prediction is obtained using majority voting among all trees.

If

$$T_1, T_2, \ldots, T_n$$

represent individual decision trees, then

$$Prediction = Mode(T_1, T_2, \ldots, T_n)$$

Random Forest generally produces highly accurate predictions because averaging multiple independent trees significantly reduces prediction variance.

Bagging (Bootstrap Aggregation) further improves model robustness by allowing each tree to observe different subsets of the training data.

Advantages

- Very high accuracy
- Robust against overfitting
- Handles noisy datasets effectively
- Supports high-dimensional data

Limitations

- Larger model size
- Lower interpretability compared to Logistic Regression
- Longer training time

7.4 Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron is a feed-forward Artificial Neural Network consisting of interconnected neurons organized into multiple layers.

The network consists of

- Input Layer
- Hidden Layer(s)
- Output Layer

Each neuron computes

$$y = f\left(\sum wx + b\right) \tag{6}$$

where

- w denotes learnable weights
- b represents the bias
- f is the activation function

During training, the network minimizes classification error using the backpropagation algorithm together with gradient descent optimization.

MLPs are capable of learning complex non-linear decision boundaries and often outperform traditional classifiers when sufficient training data are available.

Advantages

- Learns complex relationships
- High predictive capability
- Non-linear classification

Limitations

- Longer training time
- Requires hyperparameter tuning
- Less interpretable than traditional machine learning models

8 Experimental Setup

The experiments were performed using Python 3 in the Visual Studio Code development environment. The implementation was based on widely used open-source libraries for machine learning and Natural Language Processing.

The software libraries used during implementation are summarized in Table 3.

Table 3: Software Environment

Software	Purpose
Python 3	Programming Language
Pandas	Data Processing
NumPy	Numerical Computation
Scikit-learn	Machine Learning Models
NLTK	Text Preprocessing
Gensim	Word2Vec Embeddings
Matplotlib	Visualization
Joblib	Model Serialization

8.1 Training Configuration

The dataset was divided into training and testing subsets using an 80:20 split while maintaining the original class distribution through stratified sampling.

The following hyperparameters were used during experimentation.

Table 4: Model Hyperparameters

Model	Parameter	Value
Logistic Regression	Max Iterations	1000
KNN	Neighbors	5
Random Forest	Trees	200
Random Forest	Random State	42
MLP	Hidden Layers	(128,64)
MLP	Activation	ReLU
MLP	Optimizer	Adam
MLP	Epochs	20

8.2 Workflow of the Proposed System

The complete execution pipeline consists of six major stages.

1. Dataset Collection
2. Data Cleaning
3. Feature Engineering
4. Model Training
5. Performance Evaluation
6. Prediction of News Articles

9 Experimental Results

The performance of the proposed fake news detection system was evaluated using four supervised machine learning algorithms. Each model was tested on the unseen testing dataset generated after the 80:20 train-test split. The evaluation was performed using four standard classification metrics, namely Accuracy, Precision, Recall, and F1-Score. In addition, confusion matrices were generated for each classifier to analyze prediction behavior.

The objective of the experimental evaluation was not only to determine the most accurate classifier but also to compare different machine learning paradigms under identical preprocessing and feature extraction conditions.

9.1 Evaluation Metrics

Four widely accepted evaluation metrics were used to measure classification performance.

9.1.1 Accuracy

Accuracy measures the overall percentage of correctly classified news articles.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

where

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

9.1.2 Precision

Precision measures how many predicted positive samples are actually positive.

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

High precision indicates that the classifier rarely labels fake news incorrectly as genuine.

9.1.3 Recall

Recall represents the ability of the classifier to correctly identify all positive samples.

$$Recall = \frac{TP}{TP + FN} \tag{9}$$

A high recall indicates that very few genuine news articles are missed during classification.

9.1.4 F1-Score

The F1-Score is the harmonic mean of Precision and Recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{10}$$

It provides a balanced performance measure, especially when dealing with classification problems.

9.2 Performance Comparison

Table 5 summarizes the experimental performance of all four machine learning algorithms.

Table 5: Performance Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	98.74%	98.26%	99.11%	98.69%
KNN	97.24%	95.74%	98.60%	97.15%
Random Forest	**99.79%**	**99.67%**	**99.88%**	**99.78%**
Neural Network	99.00%	99.06%	98.83%	98.95%

From Table 5, Random Forest achieved the highest performance across all evaluation metrics. The model obtained an accuracy of 99.79%, precision of 99.67%, recall of 99.88%, and F1-score of 99.78%, outperforming all remaining classifiers.

The Neural Network also demonstrated excellent classification capability with an overall accuracy of 99.00%, indicating that neural architectures can effectively model textual representations extracted through TF-IDF.

Logistic Regression achieved an accuracy of 98.74%, confirming that linear models remain highly competitive for sparse textual feature spaces.

K-Nearest Neighbors produced comparatively lower performance, although its classification accuracy of 97.24% still demonstrates good predictive capability.

9.3 Confusion Matrix Analysis

The confusion matrix provides a detailed visualization of prediction outcomes by comparing actual class labels against predicted labels.

A perfect classifier would contain values only along the principal diagonal, indicating correct predictions for every sample.

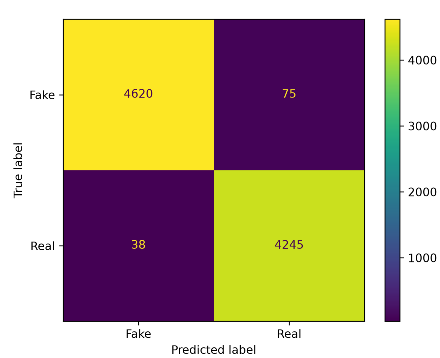

Figure 1: Confusion Matrix for Logistic Regression

The Logistic Regression classifier correctly classified the majority of fake and genuine news articles with only a small number of misclassifications, demonstrating excellent linear separability after TF-IDF feature extraction.

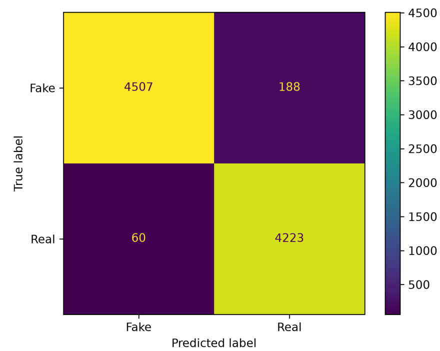

Figure 2: Confusion Matrix for K-Nearest Neighbors

The KNN classifier exhibited slightly larger off-diagonal values compared to the remaining classifiers, indicating relatively higher false positive and false negative predictions. Nevertheless, overall classification performance remained satisfactory.

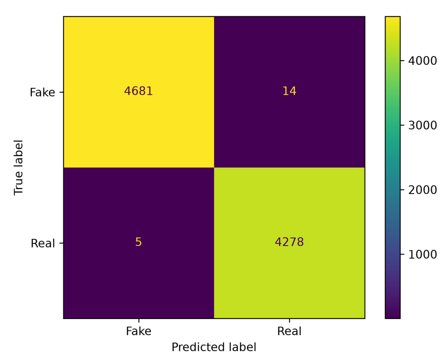

Figure 3: Confusion Matrix for Random Forest

The Random Forest confusion matrix demonstrates nearly perfect classification performance. The diagonal values dominate the matrix, confirming the model's ability to distinguish fake and genuine news articles with extremely high accuracy.

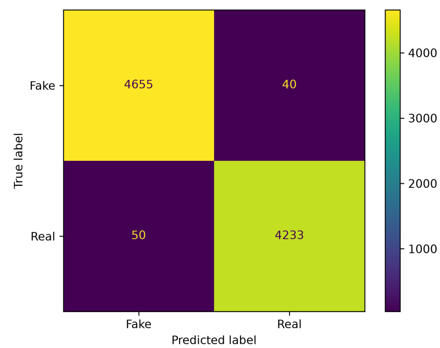

Figure 4: Confusion Matrix for Multi-Layer Perceptron

The Multi-Layer Perceptron also achieved excellent classification results. Only a limited number of samples were incorrectly classified, indicating effective learning of nonlinear decision boundaries.

9.4 Summary of Experimental Results

The experimental evaluation demonstrates that all four machine learning models achieved classification accuracies exceeding 97%, confirming the effectiveness of the adopted preprocessing and feature engineering techniques.

Among all evaluated algorithms, Random Forest consistently outperformed the remaining classifiers across every evaluation metric, making it the most suitable model for this dataset.

The obtained results further validate that carefully engineered traditional machine learning techniques remain highly competitive for fake news detection without requiring computationally expensive transformer architectures.

10 Discussion

The experimental results demonstrate that the proposed machine learning pipeline is highly effective for detecting fake news articles. Proper preprocessing of textual data, combined with suitable feature engineering techniques, significantly improved the performance of all evaluated classifiers.

One important observation from the experiments is that no single algorithm performs well solely because of its complexity. Instead, the quality of preprocessing and feature extraction plays a major role in determining classification accuracy.

The cleaning process removed unnecessary information such as URLs, HTML tags, punctuation, numbers, and stopwords, allowing the models to learn meaningful textual patterns rather than irrelevant noise. Furthermore, converting textual information into numerical representations using Bag-of-Words, TF-

IDF, and Word2Vec enabled the machine learning algorithms to effectively distinguish between fake and genuine news articles.

The obtained results indicate that even traditional machine learning methods can achieve excellent performance without relying on computationally intensive transformer-based architectures.

10.1 Comparison of Parametric and Non-Parametric Models

The four evaluated algorithms belong to two different categories of supervised learning models.

10.1.1 Parametric Models

Parametric models assume a predefined mathematical relationship between input features and output labels. During training, these models learn a fixed set of parameters.

The parametric algorithms evaluated in this work are:

- Logistic Regression
- Multi-Layer Perceptron (MLP)

Logistic Regression produced excellent results due to the high discriminative capability of TF-IDF features. Since TF-IDF generates sparse high-dimensional vectors, Logistic Regression efficiently identified the separating hyperplane between fake and genuine news articles.

The Multi-Layer Perceptron achieved even higher accuracy by learning nonlinear decision boundaries. Unlike Logistic Regression, MLP captures complex relationships between textual features through hidden neural layers.

10.1.2 Non-Parametric Models

Non-parametric models do not assume a predefined functional relationship between variables.

The non-parametric algorithms evaluated were

- K-Nearest Neighbors
- Random Forest

KNN performs classification by comparing distances between feature vectors. Because predictions depend on neighboring samples rather than learned parameters, prediction time increases with dataset size.

Random Forest constructs multiple decision trees using random subsets of both observations and features. Final predictions are generated through majority voting among all trees.

Among all evaluated models, Random Forest achieved the highest classification accuracy because the ensemble approach significantly reduced overfitting while improving generalization capability.

10.2 Performance Analysis

The comparative study highlights several important observations.

- Random Forest achieved the highest performance across all evaluation metrics, demonstrating excellent robustness and generalization.
- Logistic Regression performed remarkably well despite its relatively simple mathematical formulation, confirming that TF-IDF features are highly effective for textual classification.
- Multi-Layer Perceptron successfully captured nonlinear relationships and produced highly competitive performance.
- K-Nearest Neighbors achieved comparatively lower accuracy because distance-based methods are generally more sensitive to feature representation and high-dimensional datasets.

These observations suggest that ensemble learning approaches remain among the most reliable choices for fake news detection using classical machine learning techniques.

10.3 Advantages of the Proposed System

The developed fake news detection system provides several advantages.

- High classification accuracy.
- Fully automated news classification.
- Efficient preprocessing pipeline.
- Comparison of multiple feature extraction techniques.
- Comparison of multiple supervised learning algorithms.
- Easy implementation using open-source Python libraries.
- Computationally efficient compared to transformer-based models.

10.4 Limitations

Despite its excellent performance, the proposed system has several limitations.

- The dataset contains only English-language news articles.
- Binary classification does not distinguish between different types of misinformation.
- Contextual information such as publisher credibility and user behavior is not considered.

- Classical feature engineering methods may not capture deep semantic relationships as effectively as transformer models.
- Performance may decrease when evaluated on news articles from domains significantly different from the training dataset.

11 Conclusion

This project presented an Artificial Intelligence-based fake news detection system using Natural Language Processing and supervised machine learning techniques.

The proposed methodology involved dataset preprocessing, feature engineering using Bag-of-Words, TF-IDF, and Word2Vec, followed by classification using Logistic Regression, K-Nearest Neighbors, Random Forest, and Multi-Layer Perceptron.

Experimental evaluation demonstrated that all four models achieved excellent classification performance. Among them, Random Forest produced the highest overall accuracy, precision, recall, and F1-score, making it the best-performing classifier for the selected dataset.

The obtained results confirm that carefully engineered classical machine learning approaches remain highly effective for fake news detection while requiring significantly lower computational resources than modern deep learning architectures.

Overall, the developed system successfully satisfies the objectives of automatically identifying fake news articles with high accuracy and demonstrates the practical applicability of machine learning for combating online misinformation.

12 Future Scope

Several improvements can further enhance the proposed fake news detection system.

- Integration of transformer-based language models such as BERT and RoBERTa.
- Support for multilingual fake news detection.
- Real-time news verification using web APIs.
- Detection of misinformation shared through images and videos.
- Explainable Artificial Intelligence (XAI) techniques for improving prediction transparency.
- Deployment as a web application or browser extension for real-time news verification.
- Continuous online learning using newly collected news articles.

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